

## Biochemistry Lab #4

### Characterization and Identification of Unknown (#3) Sugar

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#### Introduction:

Carbohydrates are biological molecules that serve as the primary energy source for most organisms. They can be categorized based on their structure (mono-, di- or poly-), stereochemistry (D- or L- isomers) and on their carbonyl group (ketoses vs. aldoses). This abundance in chemical and physical properties makes it so a variety of tests are required to identify unknown sugars.

In this lab, a series of carbohydrate tests, including polarimetry, paper chromatography, Seliwanoff's test, and Nelson's test were used to identify an unknown sugar. By utilizing known standards, established physical, and chemical properties, the unknown sugar can confidently be identified.

#### Results:

The unknown #3 sugar sample was tested using paper chromatography, polarimetry, Seliwanoff's test, and Nelson's test. In Figure 1, the Unknown #3 spot aligns with the Fructose sugar spot. Polarimetry showed an initial rotation of  $-26^\circ$  and the equilibrium rotation showed  $-97^\circ$ , both of these values align with the expected values for D-fructose (Table 2). The Seliwanoff's test showed an orange color, which indicates a positive result consistent with a ketose sugar (Table 3). Lastly, the Nelson's test showed absorption values of Unknown #3 consistent with a reducing monosaccharide (Table 4).

**Table 1. Summary of Carbohydrate Tests**

| Assay Method                     | Result from Unknown #3             | Expected Result from Fructose              | Interpretation                                           |
|----------------------------------|------------------------------------|--------------------------------------------|----------------------------------------------------------|
| Paper Chromatography             | Rf and color spot matched fructose | Rf and color spot consistent with fructose | Unknown #3 and fructose arrived at the same spot         |
| Polarimetry Initial Rotation     | $-26^\circ$                        | $-21^\circ$                                | Initial rotation close to $\alpha$ -form fructose        |
| Polarimetry Equilibrium Rotation | $-97^\circ$                        | $-92^\circ$                                | Equilibrium rotation close to fructose equilibrium value |

|                   |                          |                                |                                                            |
|-------------------|--------------------------|--------------------------------|------------------------------------------------------------|
| Seliwanoff's Test | Orange (positive)        | Orange (positive)              | Confirms                                                   |
| Nelson's Test     | Strong reducing response | Strong reducing sugar response | Unknown behaves like a reducing monosaccharide (fructose). |

**Figure 1: Paper Chromatography showing Fructose and Unknown #3 Aligning**



**Table 2: Polarimetry Results for Unknown #3**

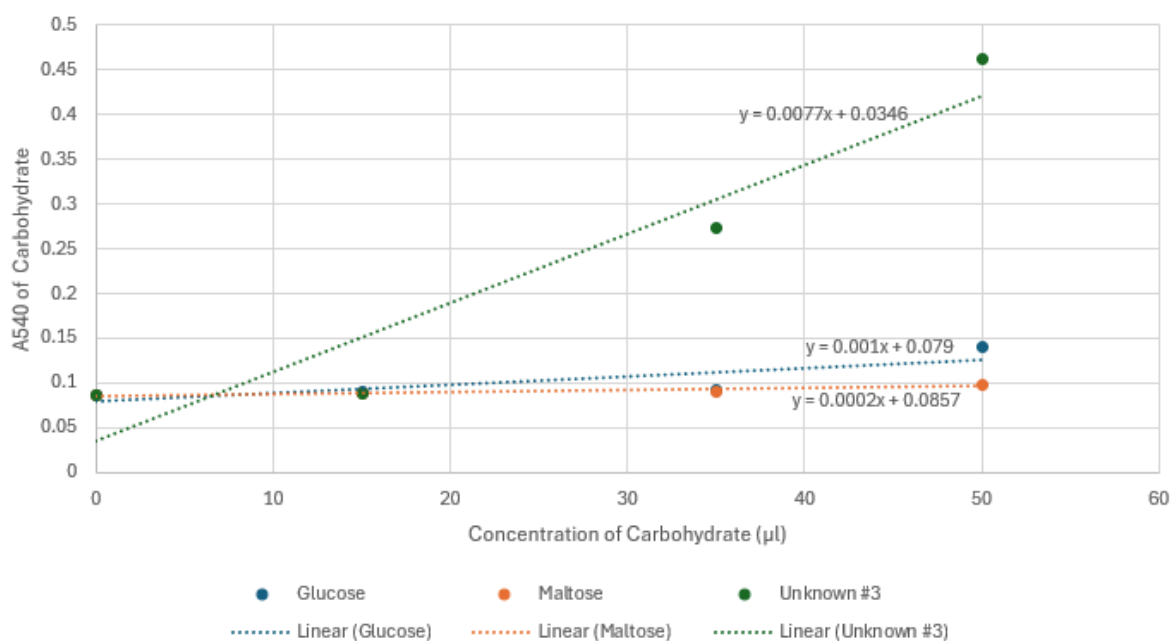
| Condition | Observed Rotation (°) | Specific Rotation (°) | Expected (°)<br>[Reference Value<br>for D-Fructose] |
|-----------|-----------------------|-----------------------|-----------------------------------------------------|
| Initial   | -4.1                  | -26                   | -21                                                 |
| Settled   | -15.5                 | -97                   | -92                                                 |

**Table 3: Results of Seliwanoff's Test**

| Sample                                 | Result |
|----------------------------------------|--------|
| 0.5 mL 0.5% glucose + 4.5mL reagent    | Clear  |
| 0.5 mL 0.5% fructose + 4.5mL reagent   | Orange |
| 0.5 mL 0.5% unknown #3 + 4.5mL reagent | Orange |
| 0.5 mL water 4.5 mL reagent            | Clear  |

**Table 4: Nelson's Test Data**

| Volume Added (mL) | Glucose A <sub>540</sub> | Maltose A <sub>540</sub> | Unknown A <sub>540</sub> |
|-------------------|--------------------------|--------------------------|--------------------------|
| 0.00              | 0.087                    | 0.087                    | 0.087                    |
| 0.15              | 0.090                    | 0.088                    | 0.089                    |
| 0.35              | 0.093                    | 0.091                    | 0.273                    |
| 0.50              | 0.141                    | 0.098                    | 0.461                    |

**Graph 1: Nelson's Test of Glucose, Maltose, and Unknown (#3) at A<sub>540</sub>**

## Discussion:

All of the test conducted in this experiment, the paper chromatography, polarimetry, Seliwanoff's test, and Nelson's test, points towards a confident identification that Unknown #3 is D-fructose. The paper chromatography (Figure 1), shows that the fructose standard and the unknown #3 spot both migrated the same distance showing similar coloration, indicating the same physical properties.

Polarimetry also confirms this conclusion, as the initial rotation of  $26^\circ$  is close to the  $\alpha$ -form of D-fructose ( $-21^\circ$ ) which is present whenever the fructose is initially dissolved into water. The equilibrium rotation of  $-97^\circ$  is also close to the reference value for the  $\beta$ -form at equilibrium ( $-92^\circ$ ) with the difference between the two measurements aligning with the expected values.

The Seliwanoff's test produced an orange color reaction, indicating that Unknown #3 is a ketose rather than an aldose. Both the fructose control and the unknown solution reacted to the heat at the same time.

Nelson's test showed a strong reduction activity for the unknown, this is consistent for a monosaccharide. The absorbance values of the glucose and maltose solutions were unexpectedly low. This suggests an issue with the concentration of these standards rather than the unknown. The unknown's strong reducing activity supports the classification of a monosaccharide (fructose).

## Materials and Methods:

The unknown (#3) sugar sample was analyzed alongside monosaccharide and disaccharide standards (see Appendix). Seliwanoff's reagent (resorcinol in 6M HCl), Nelson's Reagent (Copper (II) sulfate, sodium carbonate, and sodium potassium tartrate), a polarimeter, chromatography paper, a developing solvent, and a spectrophotometer were used in the four experiments.

To prepare the paper chromatography, small spots of 10 $\mu$ l for each standard/unknown were applied to the paper. The paper was developed in a solvent composed of 60 mL isopropanol, 20 mL Acetic Acid, and 20 mL water. The solvent was allowed to migrate towards the top of the paper using capillary action, after of which the paper was allowed to dry in a 120°C oven. After the paper was dried, the spots were developed using an aniline-oxalate spray.

For polarimetry, an 8% solution of the unknown sugar was prepared. The initial rotation was measured immediately after dissolving the sugar in water. The solution was allowed to reach

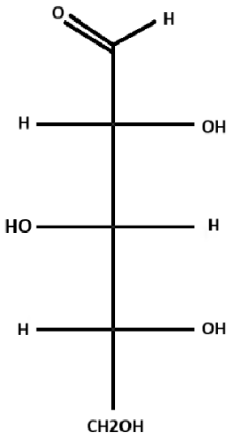
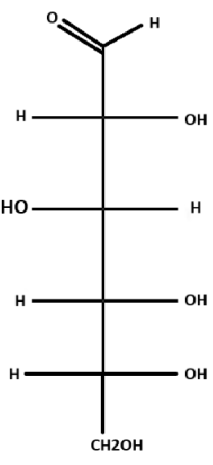
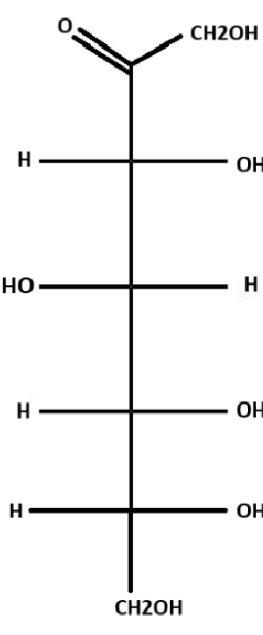
equilibrium, then another measurement was taken. These measurements were used to calculate specific rotations, then compared to the reference standard values.

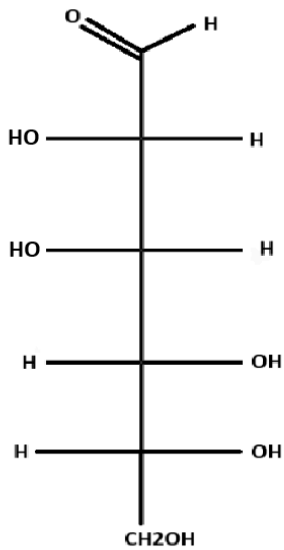
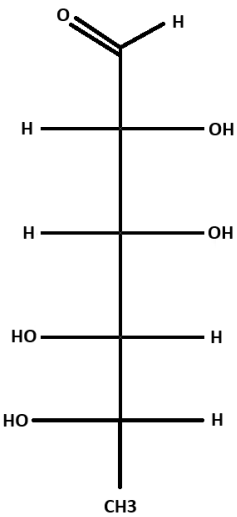
Seliwanoff's test was conducted by dissolving 0.5 mL of the unknown/sugar standards and 4.5 mL of the Seliwanoff's reagent and heated at 100°C. Development of an orange color indicates the presence of a ketose.

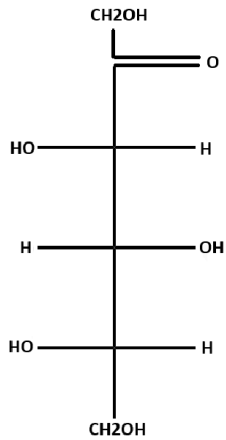
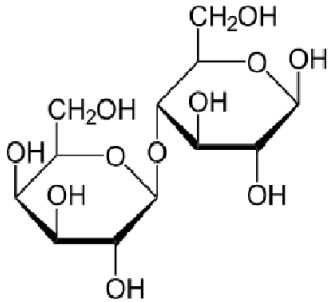
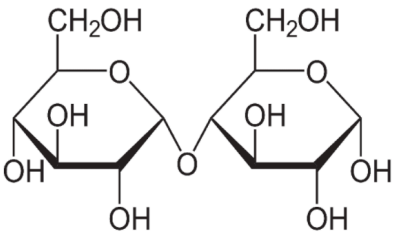
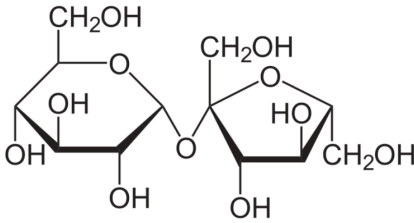
For Nelson's test, a 1mg/mL solution of the unknown solution was prepared, along with the standards as positive controls. The standards and unknown were distributed into 10 test tubes according to the lab procedure. To each tube, 0.5 mL of Nelson's Alkaline Copper Reagent was added, then the tubes were heated for 20 minutes. After cooling to room temp 0.5 mL of Arsenomolybdate Reagent was added to dissolve the precipitated  $\text{Cu}_2\text{O}$  and then diluted with 3.5 mL of water. Absorbance for each sample was measured at 540 nm then plotted.

## Appendix A: Sugar Structures

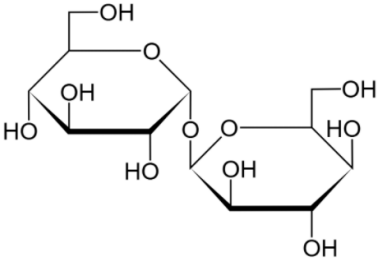
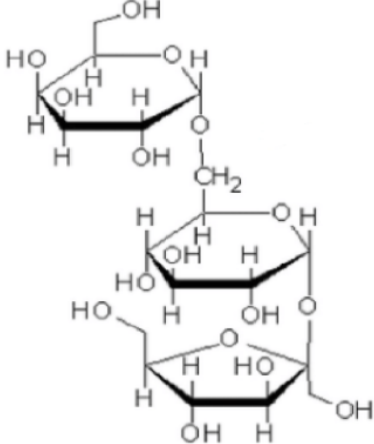
| Sugar       | Structure Type | Figure                                                                                                                                                                                                                                                       |
|-------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-Ribose    | Monosaccharide | $  \begin{array}{c}  \text{CHO} \\    \\  \text{H} - \text{C} - \text{OH} \\    \\  \text{H} - \text{C} - \text{OH} \\    \\  \text{H} - \text{C} - \text{OH} \\    \\  \text{CH}_2\text{OH}  \end{array}  $                                                 |
| L-Arabinose | Monosaccharide | $  \begin{array}{c}  \text{O} \quad \text{H} \\  \diagdown \diagup \\  \text{C} \\    \\  \text{H} - \text{C} - \text{OH} \\    \\  \text{HO} - \text{C} - \text{H} \\    \\  \text{HO} - \text{C} - \text{H} \\    \\  \text{CH}_2\text{OH}  \end{array}  $ |

|            |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-Xylose   | Monosaccharide          |  <p>The Fischer projection of D-Xylose is a five-carbon aldehyde. The top carbon (C1) is an aldehyde group with a double bond to oxygen (O) and a single bond to hydrogen (H). The second carbon (C2) has a hydrogen (H) on the left and a hydroxyl group (OH) on the right. The third carbon (C3) has a hydroxyl group (HO) on the left and a hydrogen (H) on the right. The fourth carbon (C4) has a hydrogen (H) on the left and a hydroxyl group (OH) on the right. The bottom carbon (C5) is a CH2OH group.</p>                                                                                               |
| D-Glucose  | Monosaccharide          |  <p>The Fischer projection of D-Glucose is a six-carbon aldehyde. The top carbon (C1) is an aldehyde group with a double bond to oxygen (O) and a single bond to hydrogen (H). The second carbon (C2) has a hydrogen (H) on the left and a hydroxyl group (OH) on the right. The third carbon (C3) has a hydroxyl group (HO) on the left and a hydrogen (H) on the right. The fourth carbon (C4) has a hydrogen (H) on the left and a hydroxyl group (OH) on the right. The fifth carbon (C5) has a hydrogen (H) on the left and a hydroxyl group (OH) on the right. The bottom carbon (C6) is a CH2OH group.</p> |
| D-Fructose | Monosaccharide (ketose) |  <p>The Fischer projection of D-Fructose is a six-carbon ketose. The top carbon (C2) is a ketone group with a double bond to oxygen (O) and single bonds to a hydrogen (H) on the left and a CH2OH group on the right. The third carbon (C3) has a hydrogen (H) on the left and a hydroxyl group (OH) on the right. The fourth carbon (C4) has a hydroxyl group (HO) on the left and a hydrogen (H) on the right. The fifth carbon (C5) has a hydrogen (H) on the left and a hydroxyl group (OH) on the right. The bottom carbon (C6) is a CH2OH group.</p>                                                     |

|            |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-Mannose  | Monosaccharide |  <p>The Fischer projection of D-Mannose is a five-carbon aldehyde. The top carbon (C1) is an aldehyde group with a double bond to oxygen (O) and a single bond to hydrogen (H). The second carbon (C2) has a hydroxyl group (HO) on the left and a hydrogen atom (H) on the right. The third carbon (C3) has a hydroxyl group (HO) on the left and a hydrogen atom (H) on the right. The fourth carbon (C4) has a hydrogen atom (H) on the left and a hydroxyl group (OH) on the right. The fifth carbon (C5) has a hydrogen atom (H) on the left and a hydroxyl group (OH) on the right, and is bonded to a CH2OH group at the bottom.</p> |
| L-Rhamnose | Monosaccharide |  <p>The Fischer projection of L-Rhamnose is a five-carbon aldehyde. The top carbon (C1) is an aldehyde group with a double bond to oxygen (O) and a single bond to hydrogen (H). The second carbon (C2) has a hydrogen atom (H) on the left and a hydroxyl group (OH) on the right. The third carbon (C3) has a hydrogen atom (H) on the left and a hydroxyl group (OH) on the right. The fourth carbon (C4) has a hydroxyl group (HO) on the left and a hydrogen atom (H) on the right. The fifth carbon (C5) has a hydroxyl group (HO) on the left and a hydrogen atom (H) on the right, and is bonded to a CH3 group at the bottom.</p> |

|           |                            |                                                                                      |
|-----------|----------------------------|--------------------------------------------------------------------------------------|
| L-Sorbose | Monosaccharide<br>(ketose) |    |
| Lactose   | Disaccharide               |   |
| Maltose   | Disaccharide               |  |
| Sucrose   | Disaccharide               |  |



|           |               |                                                                                     |
|-----------|---------------|-------------------------------------------------------------------------------------|
| Trehalose | Disaccharide  |   |
| Raffinose | Trisaccharide |  |

Haworth Projections from Wikipedia Commons (Lactose, Maltose, Sucrose, Trehalose, and Raffinose)

## Appendix B: Polarimetry Reference

| Sugar       | Alpha form | Equil. mixture | Beta form |
|-------------|------------|----------------|-----------|
| D-ribose    | -23.1      | -23.7          | --        |
| L-arabinose | +54.0      | +104.5         | +175.0    |
| D-xylose    | +92.0      | +19.0          | -20.0*    |
| D-glucose   | +113.4     | +52.2          | +19.0     |
| D-galactose | +144.0     | +80.5          | +52.0     |
| D-fructose  | -21.0*     | -92.0          | -133.5    |
| D-mannose   | +34.0      | +14.6          | -17.0     |
| L-rhamnose  | -7.7       | +8.9           | +54.0*    |
| L-sorbose   | --         | -43.4          | --        |
| lactose     | +90.0      | +55.3          | +35.0     |
| maltose     | +168.0*    | +136.0         | +118.0    |
| Sucrose     | --         | +66.5          | --        |
| Trehalose   | --         | +178.3         | --        |
| Raffinose   | --         | +105.2         | --        |

\*calculated value.